Production scheduling and equipment system selection

By

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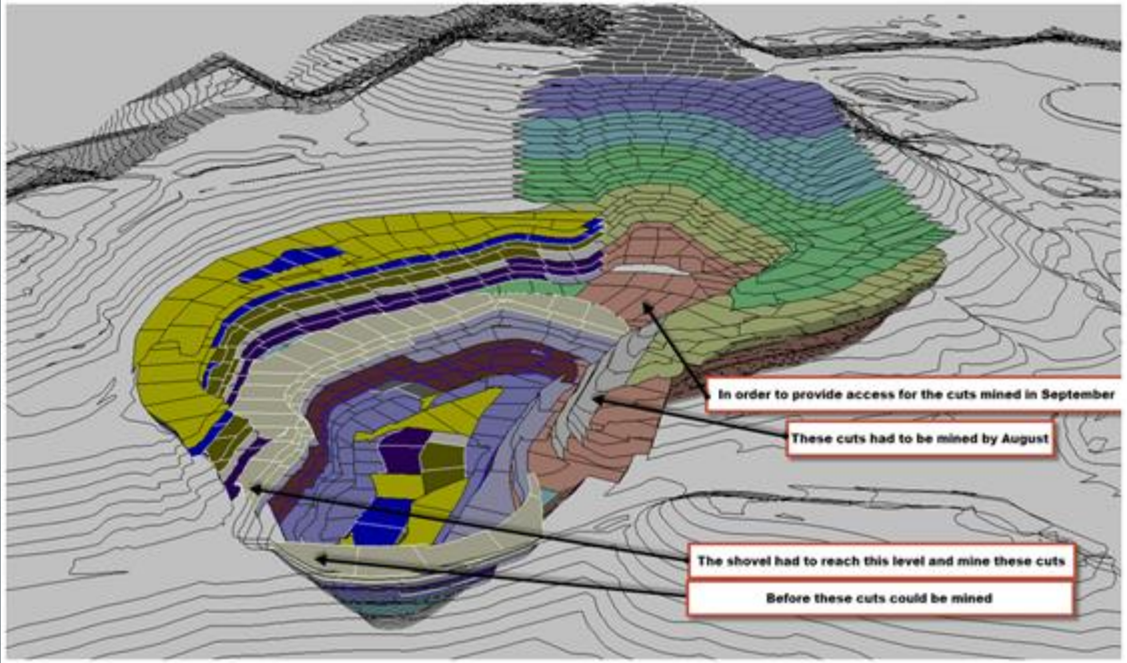
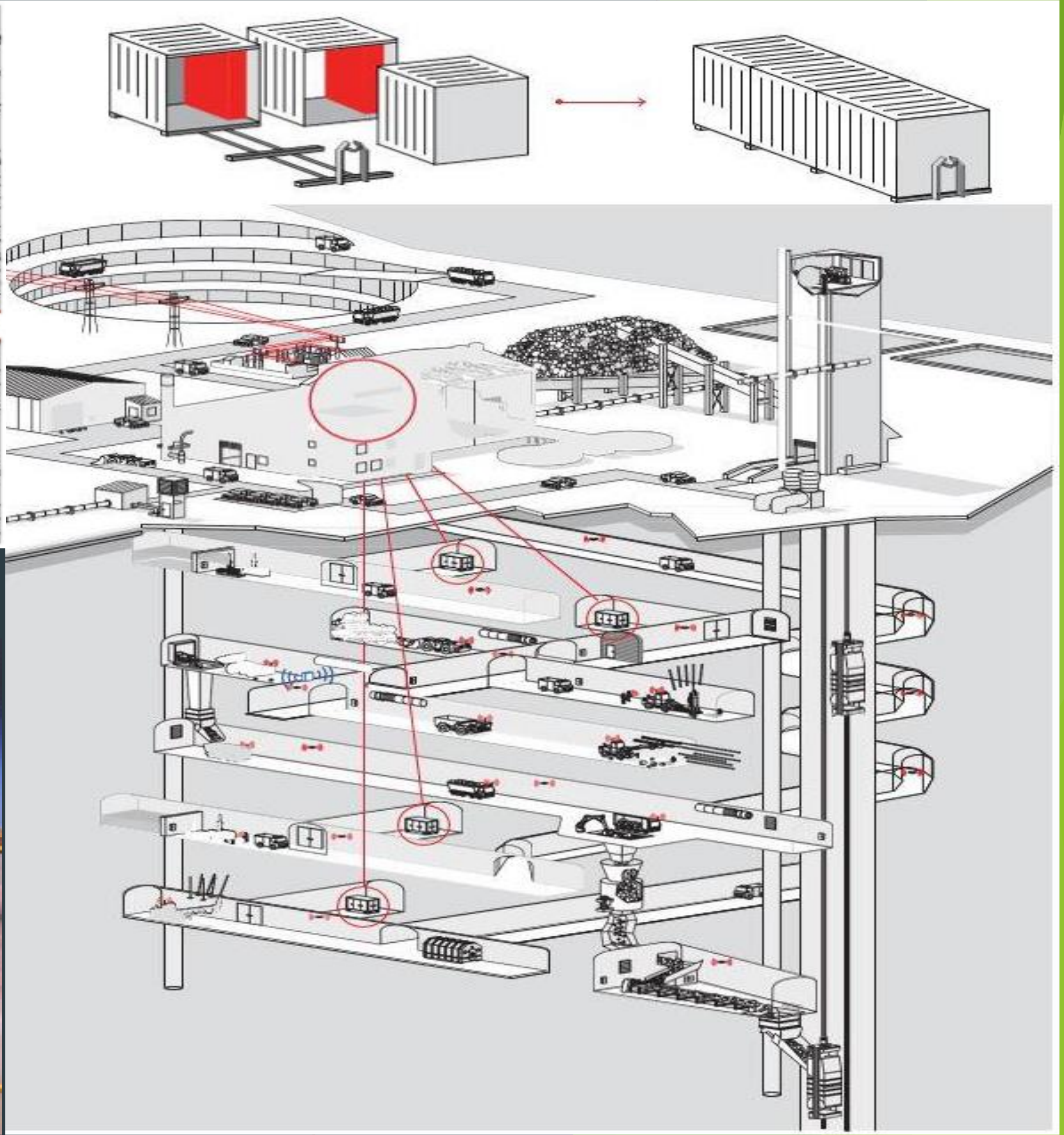
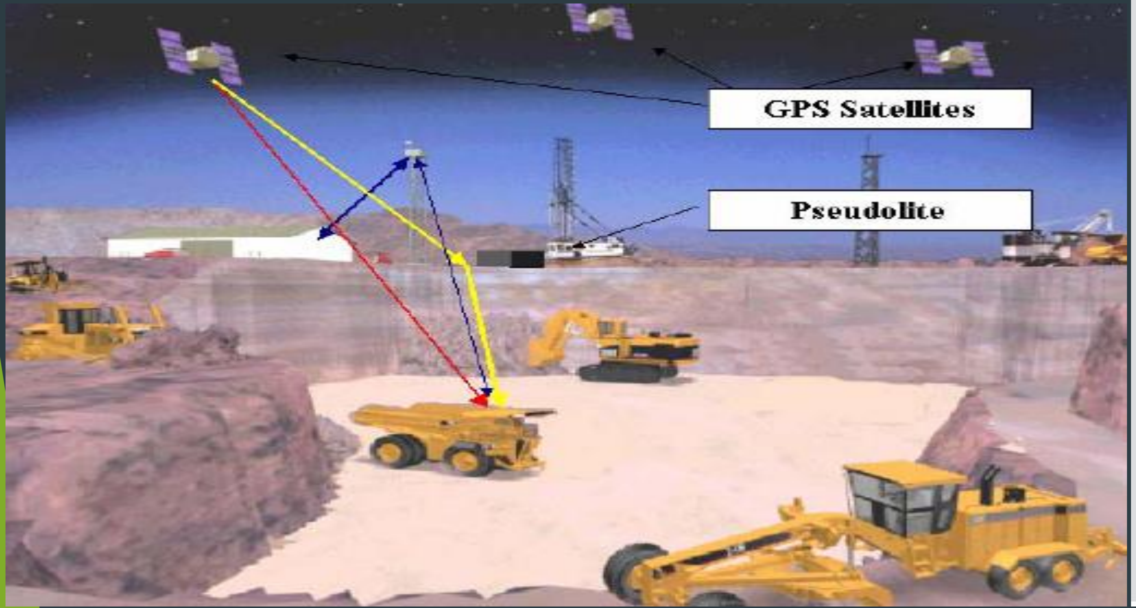
Production scheduling and equipment system selection

- Production scheduling concepts,
- Optimum mine size and Taylor’s mine life rule,
- Selection of equipment system and scheduling


In order to provide access for the cuts mined in September
These cuts had to be mined by August
The shovel had to reach this level and mine these cuts
Before these cuts could be mined




GPS Satellites
Pseudolite

Introduction

Production scheduling aims to define the most profitable extraction sequence of the mineralized material from the ground that produces maximum possible discounted profit while satisfying a set of physical and operational constraints.

the two key problems in production scheduling are "priorities" and "capacity." In other words, "What should be done first?" and "Who should do it?" Wight (1984) defines scheduling as "establishing the timing for performing a task" and observes that there are multiple types of scheduling. Cox et al. (1992) define detailed scheduling as "the actual assignment of starting and/ or completion dates to operations or groups of operations to show when these must be done if order is to be completed on time. “They note that this is also known as operations scheduling, order scheduling, and shop Scheduling".

It is also possible to estimate any changes in production capacity to achieve the target that can be feasible. The effectiveness of mine scheduling will influence the long term and short term performance of a mine. Production scheduling is an important face of mine planning. Initial production scheduling is based on marginal analysis of ore reserves, a haulage study based on a conceptual pit design and overall facility layout

A Gantt chart is "the earliest and best known type of control chart especially designed to show graphically the relationship between planned performance and actual performance." Gantt designed his charts so that foremen or other supervisors could quickly know whether production was on schedule, ahead of schedule, or behind schedule. Modern project management software includes this critical function even now.

Production scheduling, along with production planning, provides projections of future mining progress and time requirements for the development and extraction of a resource. These schedules and plans are used by management as a means of attaining the following objectives: (1) maintaining or maximizing expected profit, (2) determining future investment in mining, (3) optimizing return on investment, (4) evaluating alternative investments, and (5) conserving and developing owned resources.

Production scheduling is important to the overall mine design because of he substantial costs associated with labor, supplies, and equipment which are affected by the production schedule.

The production schedule is a plan relating to (1) production rate and (2) operating layout. These factors establish the main criteria for the development of a production schedule. The production rate determines the limits of production capacity for a production unit such as a shovel and a fleet of haulage trucks.
A series of these production units establishes the overall production of the mine. The operating layout establishes the physical constraints which will be encountered by the production units. Time, a finite constraint, establishes the duration or length of the schedules.

The production rate is material per unit of time for an equipment unit or a series of equipment units. The material factor of he production rate can be described as follows: (1) metric tons (short tons) per hour, shift, day or year, and (2) cubic meters (cubic yards) per hour, shift, day or year.

The operating layout element of production scheduling is the establishment of the physical or operating constraints of the mine design. Some of the key factors that must be taken into account when developing an operating layout are: (1) established pit operating procedures, (2) expected ore grades, (3) planned operating slopes, (4) designed haul roads, (5) planned dump development, (6) planned backfilling and reclamation sequences, (7) designed surface and ground-water controls, (8) required equipment size and maneuverability, and (9) planned bench development.

Production scheduling also analyzes the production capacity of the existing equipment. This evaluation usually includes an examination of previous equipment performance levels and a projection of expected equipment performance. The major items reviewed by the engineer are: (1) expected equipment fleet sizes; (2) project equipment availability; (3) projected equipment utilization; (4) planned haulage profiles and conditions; and (5) anticipated mining conditions, digging, development work, weather, water, and labor constraints. These items allow the engineer to schedule or adjust a short-range mine plan and to develop a production plan which the mine operator can use in meeting operating goals and objectives.

Methods for Production Scheduling

- Manual and computer methods.

The application of graphic digitizers used in conjunction with the computer has provided the tools necessary to handle large masses of data for defining individual shovel cuts and cut sequences as well as shift or daily ore grade control. This production planning system is presently being used to: (1) forecast pit area geometry by shovel cut, (2) display ore grades by cut and to guide how to blend ores for grade control, (3) prepare production schedules, and (4) compare mining development progress to production schedules.

Another major computer method used in production planning is the truck simulation model. To schedule and evaluate haul truck requirements properly, computerized truck simulation models have been developed and are used almost universally by mining companies and equipment manufacturers. This simulation model is based on a stochastic process based on the performance curve supplied by the equipment manufacturer. Stochastic simulation is particularly valuable when analyzing the effect of scheduling trucks of significantly different size or speed to meet a desired production rate. Although this technique has been used extensively for haul trucks, it is being used for scrapers, trams, and highway trucks.

Introduction

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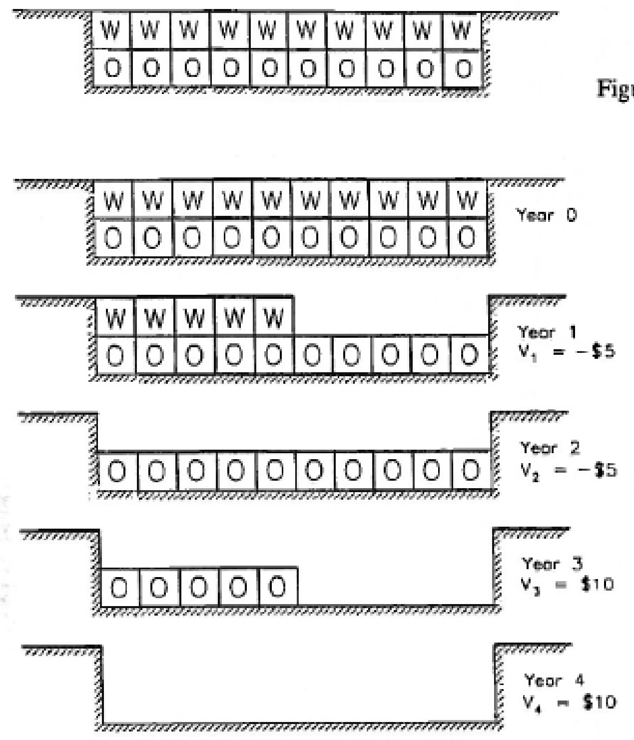


Figure 6.28. Simple sequencing example.

Figure 6.29. Mining sequence scenario 1.

Scenario 1. Removal of waste followed by ore mining

For operational simplicity it would be best if all of the waste could be stripped (removed) first followed later by ore mining.

The net present value for this sequence assuming an interest rate of 10% is

$$\text{NPV} = \frac{-\$5}{(1.10)^1} + \frac{-\$5}{(1.10)^2} + \frac{\$10}{(1.10)^3} + \frac{\$10}{(1.10)^4}$$
$$= -\$4.55 - \$4.13 + \$7.51 + \$6.83 = \$5.66$$

Scenario 2. One year of pre-stiipping followed by both ore (3 blocks/year) and waste (2 blocks/year) mining

This alternative would require mining 5 blocks of waste in year 1. In years 2 and 3 three blocks of ore would be mined for every two blocks of waste. The final year would have 1 block of waste and 4 blocks of ore.

The net present value is now

$$\mathrm{NPV} = \frac{-\$5}{(1.10)^1} + \frac{\$4}{(1.10)^2} + \frac{\$4}{(1.10)^3} + \frac{\$7}{(1.10)^4}$$
$$= -\$4.54 + \$3.31 + \$3.01 + \$4.78 = \$6.56$$

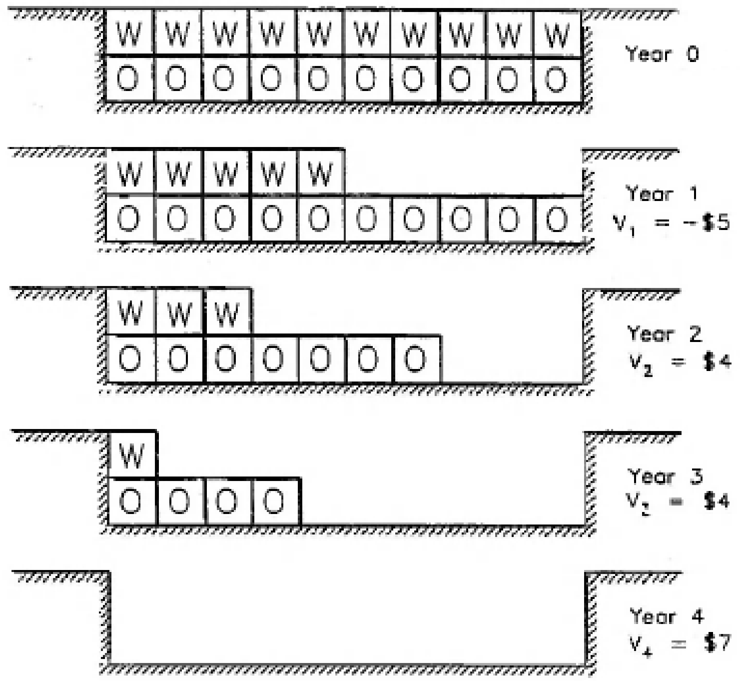


Figure 6.30. Mining sequence – scenario 2.

Scenario 3. Mining of waste is maintained one block ahead of ore Comparing scenarios 1 and 2, there was an improvement in the net present value when the time lag between stripping and mining was shortened. In this third scenario, the stripping will be kept only one block ahead of ore mining, to make the time lag even shorter.

$$\text{NPV} = \frac{\$1}{(1.10)^1} + \frac{\$2.50}{(1.10)^2} + \frac{\$2.50}{(1.10)^3} + \frac{\$4}{(1.10)^4}$$
$$= \$0.91 + \$2.07 + \$1.88 + \$2.73 = \$7.59$$

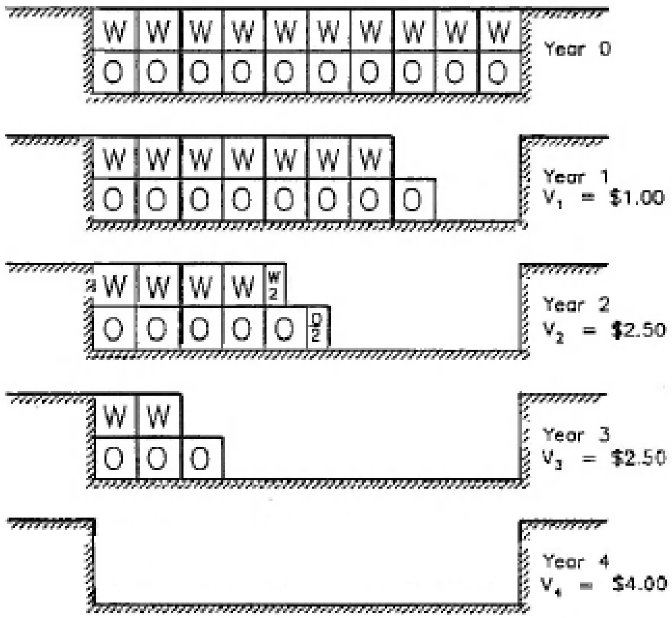


Figure 6.31. Mining sequence – scenario 3.

Scenario 4. Stripping is maintained one half block ahead of ore mining A major improvement in the NPV was observed between scenarios 2 and 3. To explore this further, consider the situation when the stripping lead is cut to one-half block.

The NPV is

$$NPV = \frac{\$1.75}{(1.10)^1} + \frac{\$2.50}{(1.10)^2} + \frac{\$2.50}{(1.10)^3} + \frac{\$3.25}{(1.10)^4}$$
$$= \$1.59 + \$2.07 + \$1.88 + \$2.22 = \$7.76$$

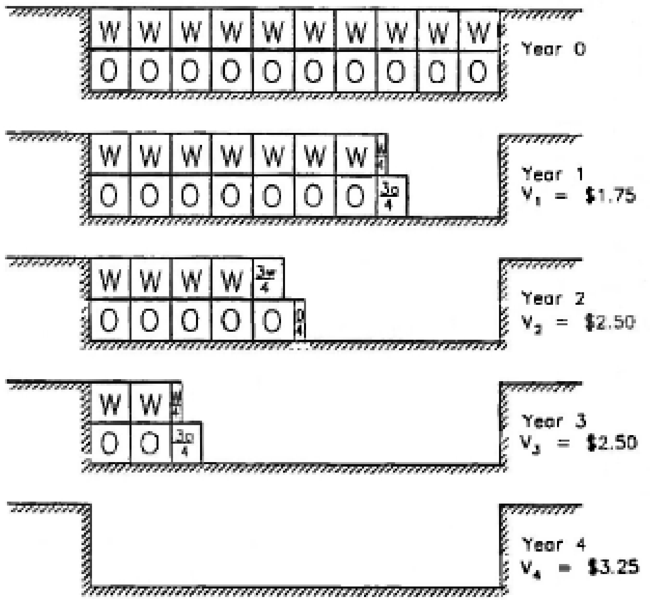


Figure 6.32. Mining sequence –scenario 4.

Scenario 5. Mining rate doubled

It has been suggested that with the purchase of more equipment, the mining rate could be increased to 10 blocks per year. There would be an increase in the equipment ownership costs to be charged against both ore and waste however.

The resulting values are:

Waste cost = \$1.10/block

Ore revenue = \$1.90/block

Stripping will be kept one block ahead of ore mining as in

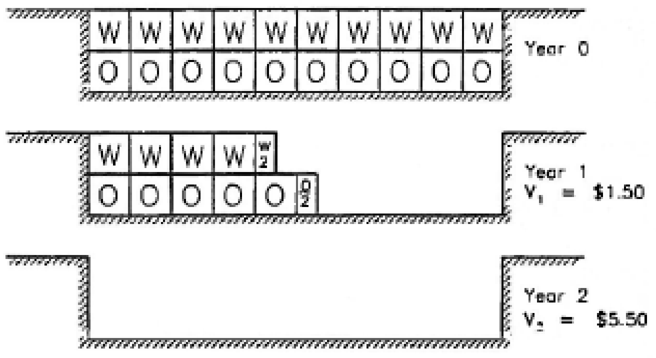


Figure 6.33. Mining sequence – scenario 5.

The NPV is

$$NPV = \frac{\$2.50}{(1.10)^1} + \frac{\$5.50}{(1.10)^2} = \$2.27 + \$4.55 = \$6.82$$

As can be seen scenario 3 remains the most favorable.

If there had been no additional cost then

$$NPV = \frac{\$3.50}{(1.10)^1} + \frac{\$6.50}{(1.10)^2} = \$3.18 + \$5.37 = \$8.55$$

and scenario 5 would have been the most favorable.

The NPV is dependent upon

1. The time interval between stripping and ore mining. It is highest when the lead time is short. With added costs associated with shortening the lead time, however, there may or may not be an improvement in NPV.
2. The production rate. For the same unit cost, the highest NPV is achieved with the highest production rate. With added costs with increasing production rate, there may or may not be an improvement in NPV.

Phase scheduling

- Several mining areas or mine phases, typically three or more, are active at any given time during the life of a mine. Of these, one or two would be in the process of being stripped, another being mined for ore and the last nearing exhaustion. This section will describe a procedure which can be used to help sequence the phases so that the desired ore stream is produced. The planning scheduling will be described in a step-by-step manner.
- The phases are first designed. The slope angles used are selected based upon initial geotechnical investigations. For this example it is assumed that the orebody is of uniform grade. Hence the phases A through F (shown in Figure) have been designed, subject to access constraints, to progressively mine the 'next best' ore in terms of annual stripping ratio.
- The ore-waste tonnage inventory by bench and phase is determined. At this point the mining engineer would 'mine' the ore on each successive bench gradually stepping, in order, through the phases to meet the required annual mill production. Such an ore schedule would typically be done using a hand calculator or an interactive desk top computer program. Each phase would have a different ore life since they were defined on operating rather than schedule constraints. The first trials would be based upon a fixed cutoff grade.

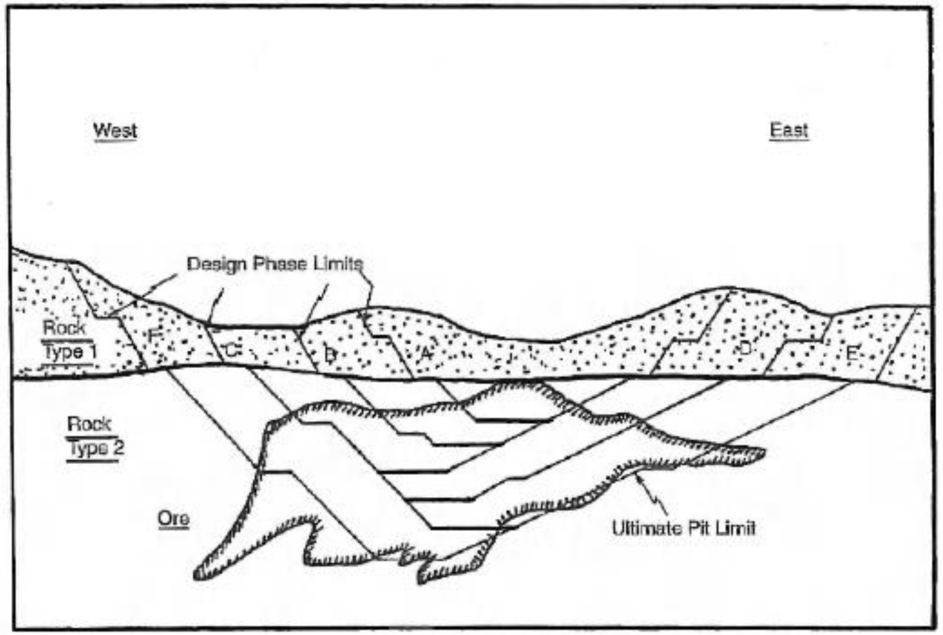


Figure 6.35. Development scheme for the hypothetical deposit (Mathieson, 1982).

Concluding Remarks

Mine production planning and production scheduling in surface mines are basically iterative processes. The aim of mine production scheduling is to generate a plan or schedule of selected part of an ore body that will ensure uniform grade and targeted tonnage based on the plant requirements.

Production scheduling has also an objective to schedule the production process so that it is able to improve the production adherence and also to improve the efficiency.

Today production scheduling tools are highly based on powerful graphical interface and it enables solutions which can be used to optimize work load in relative basis at every stage of production.

Production scheduling in the mine can be done either forward or backward scheduling to allocate plant, machinery and manpower resources and also to schedule the production and purchase inventories. In the case of forward scheduling, the various tasks are planned starting from the date where the resources become available and the date for shipping the ore determined. In the case of backward scheduling, the various tasks are planned so that the starting and completion date of each task is determined to ship the mineral.

The traditional research on production scheduling in mines has been based on the development of a sequence of resources depletion schedules leading from the initial condition of the ore deposit to the ultimate pit limits.

The production scheduling problem is characterized by the duration of scheduling period.

- Long term production scheduling is generally done on yearly basis and usually five years. This gives information for strategic mine planning, planning ore reserve, major investment planning and stripping ratio.
- Short term production scheduling is done on daily, weekly, monthly or quarterly or even yearly basis. The operation scheduling is done to provide required grade and tonnage of mineral, equipment positioning and operation.
- Medium term production scheduling is done for short range planning of mining operation, budgeting and inventory requirements.

Operations need to be carried out while undertaking surface mining

Site preparation
(Removal of vegetation)
↓
Cleaning & grubbing
↓
Top soil removal
(Transport to store it or for the direct replacement)
↓
Rip, scrap, doze ground i.e. over burden
(Depending upon type of ground encountered,
which could be unconsolidated, consolidated etc.)
↓
Put box-cut or create initial free face in o/b or waste rock
↓
Drive benches in o/b or waste rock applying unit operations
(Drilling, blasting, mucking, transportation whatever applicable)
↓
Put box cut or create initial free face in orebody to be mined
↓
Drive benches in orebody
↓
Rock fragmentation by
↓
Drilling & blasting | Ore cutting | Ripping
↓
Ore excavation/loading/mucking/hoisting
↓
Ore transportation
↓
Removal of parting of waste rock, if any
and mining the rest of the orebody
↓
Back filling in case of open cast mines
↓
Spoil regrading
↓
Top soil replacement & final grading
↓
Revegetation
↓
Support activities such as: drainage, access roads, maintenance and environment
management to be carried simultaneously

Optimum mine size and Taylor’s mine life rule

- Proper estimation of targeted production vis-à-vis mine life for a given deposit is of paramount importance. Too low production rate sacrifices possible economics of scale and defers possible profits too far into the future. Conversely, too high a production rate may drive up the project’s capital cost beyond any ability to repay within the shortened mine life.
- Taylor (1977), after studying many actual projects involving a wide range of ore body sizes and shapes for which the total ore reserves were reasonably well known, put forward an empirical relationship (Taylor’s mine life rule) between the total ore tonnage and the mine life. The rule, a simple and useful guide, states:
- Life (years) = 0.2 x (Expected ore tonnage)^1/4
- However, it is more convenient to use quantities expressed in millions, and generally the practical range of variation of mine life seems to lie within □ 20%. So the rule may be restated as:
- Life (years) = (1 +/- 0.2) x 6.5 x (Tonnage in millions)^1/4

- The size of a mine, which defines the rate of mine production, is one of the most important variables affecting the viability of the mining project . The selection and sizing of mining and milling equipment as well as the estimation of manpower requirements are based mainly on the production rate of the mine. Accordingly, the initial capital outlays and the annual operating expenses are directly influenced by the selected production rate.
- The bigger the proposed mine the greater will be the capital investment, and the greater the annual revenues, but for a shorter period of time. The smaller the mine, the smaller the investment and the smaller the annual cash flow, but for a longer period of time. Both of the extremes influence the economic success of the project. If the production rate is too low, the annual net cash flows may be insufficient to repay the initial capital investment at the defined discount rate. If the production rate is too high, the initial capital investment required may undermine the project economics, especially when the mine size exceeds the infrastructure capacity of the region. Also, over-sizing of a project imposes a greater risk of failure, with less time to recover from a bad start

- Wells (1978) outlined a method for optimizing the production rate of a mining project . The selection of the optimum rate is carried out using the present value ratio (PVR) as the optimizing criterion. The PVR is defined as follows:

PVR = PVOUT/PVIN

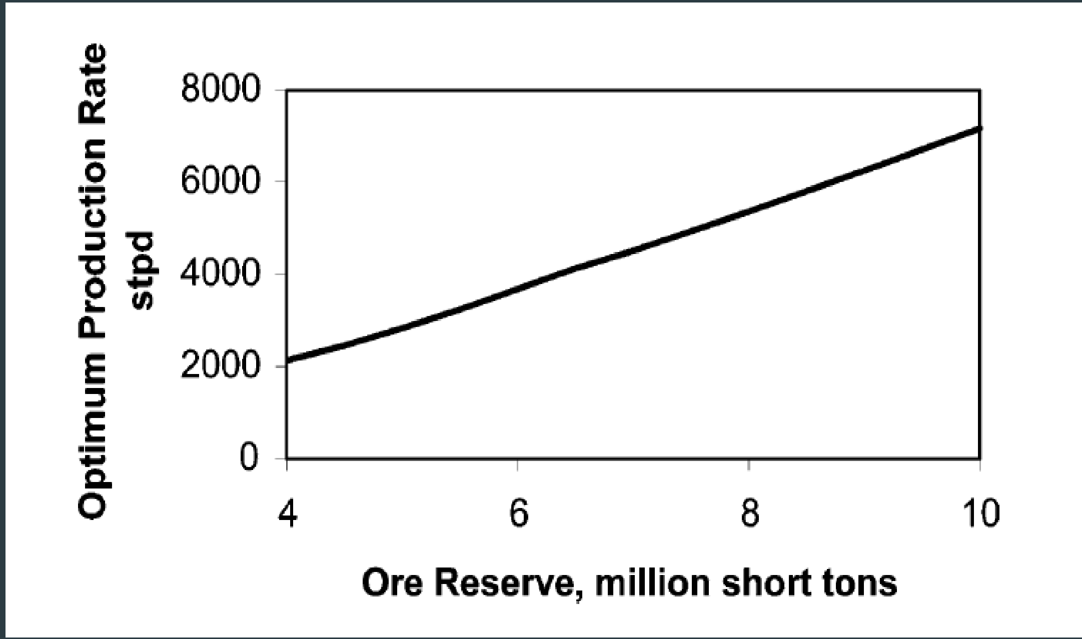
- Where PVOUT is the present value of positive cash flows and PVIN is the present value of negative cash flows. A PVR < 1 indicates an unsatisfactory return on investment; a PVR =1 indicates an acceptable return on the investment; and a PVR > 1 indicates a return in excess of the minimum requirement. The optimum production rate by this method is the one that yields the maximum PVR.

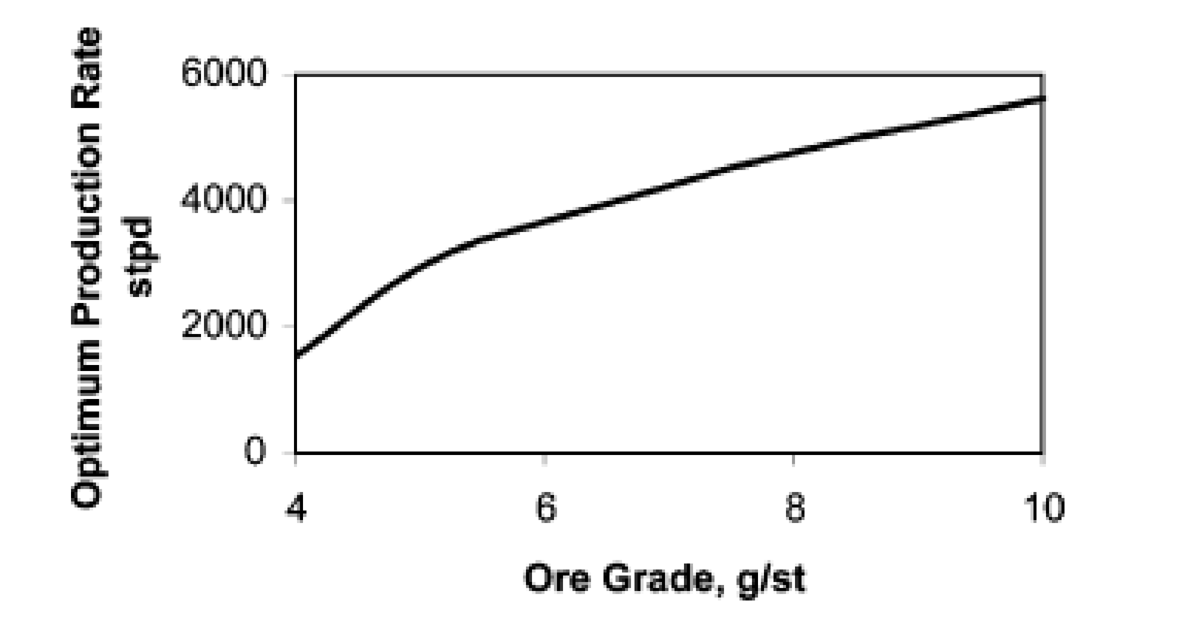
Factors affecting the selection of the optimum production rate

- Factors related to the physical characteristics of the deposit,
- Economic factors, and
- Financial factors.

The physical characteristics of the deposit

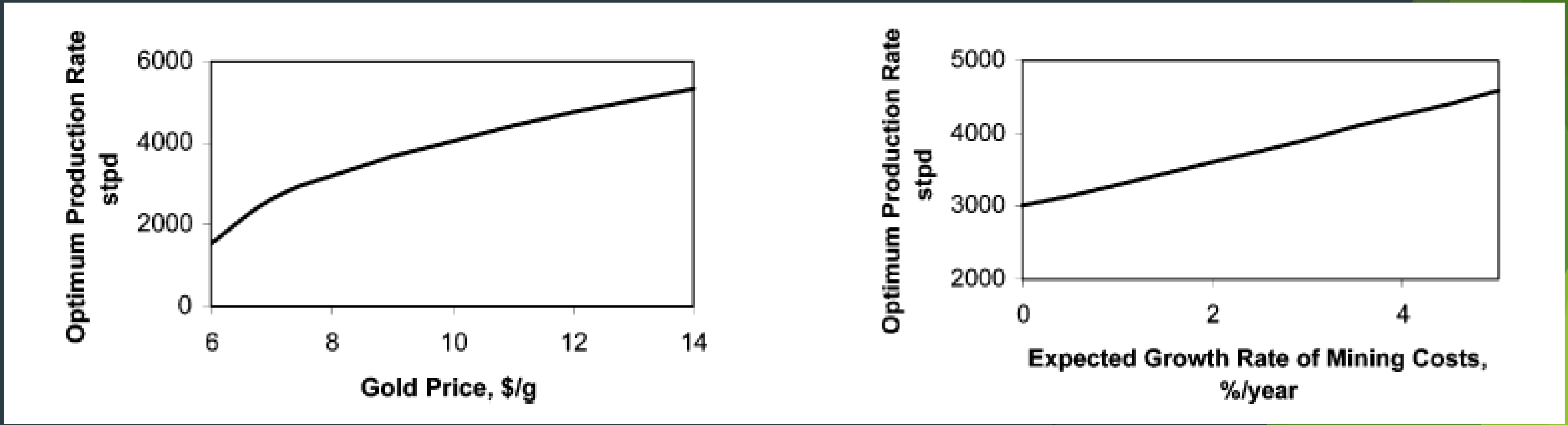
- Both the reserve tonnage and the ore grade affect the expected future revenues from the mining project, and hence they affect the optimum size of the mine. As shown in Fig. as the amount of recoverable ore reserve increases, the optimum production rate increases





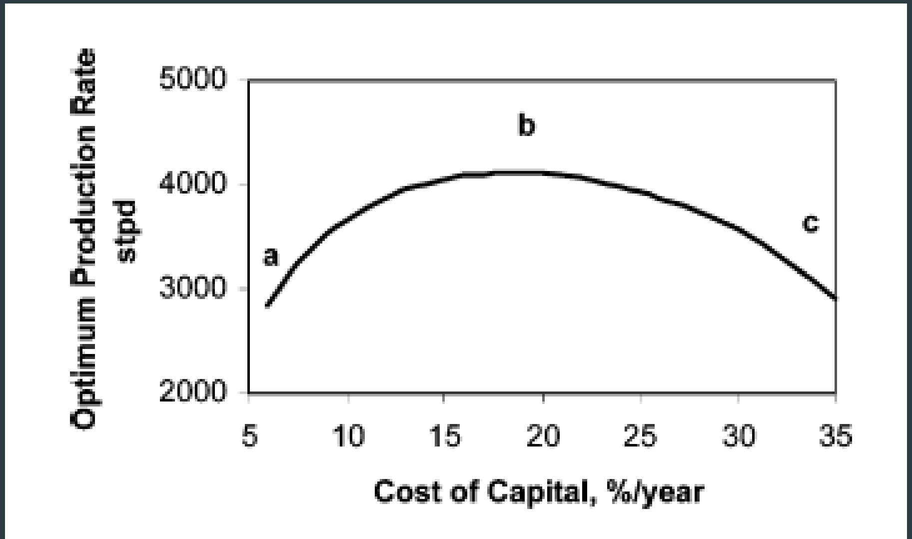
Economic factors

- Among the more important economic factors affecting the optimum mine size are: the price of the produced mineral, the expected rate of change of the mineral price, and the expected rate of change of the mining costs due to inflation or other factors.
- The price of the produced mineral directly affects the profit per ton of the ore mined. As the mineral price increases, the profit per ton increases and thus the optimum production rate increases
- The mineral price throughout the mine life is not constant but it changes from year to year. It is important to take into account the expected rate of change of the mineral price and discuss its effect on the optimum production rate. As



Financial factors

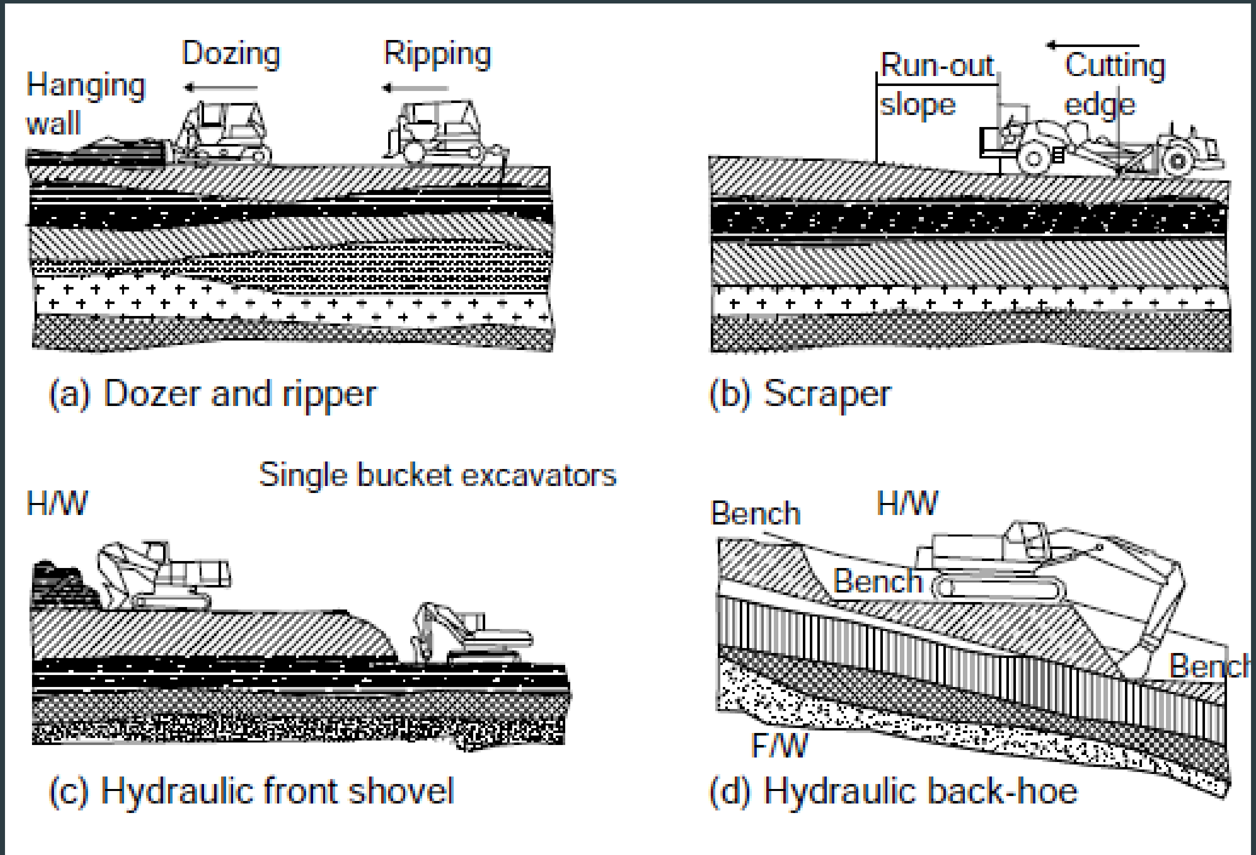
- Companies usually raise funds to finance projects by different ways including, most importantly, debt, equity, and internal finance. The cash flows generated from a project should be sufficient to repay the borrowed capital, pay the interest, and produce a net profit. The financial decision of a company determines its capital structure, which in turn determines the company's cost of capital. A company's cost of capital is the one that should be used to discount back future cash flows to the present, and hence it affects the investment decision as well as the planning policy of the company. Since the optimum size of the mine depends on the present values of both revenues and costs, it is greatly influenced by the financial decision in the form of the cost of capital.

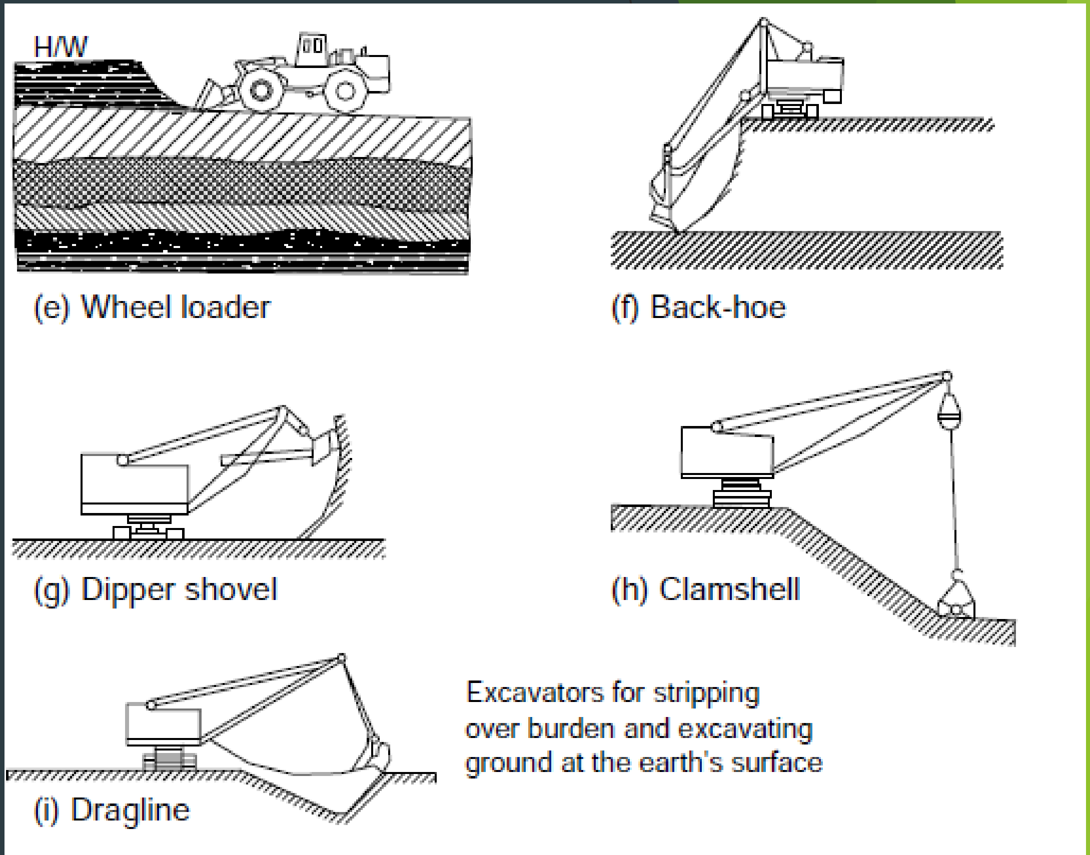


- The production rate (or size) of a mining project is the basic determinant of the capital and operating costs as well as the revenues generated from the project

Mine size : Bankura Rev Cost 480 Option.xls

Selection of equipment system and scheduling





Excavators used for earth-work, rock extraction and ore mining.

Planning the equipment system

- The surface mining being a machine intensive one, the economic success of a surface mining project depends to a great extent on the selection of the right equipment system in relation to the chosen mining system and the method of work.
- Factors that affect the choice of equipment in a mine are: (i) required production, (ii) haul /transport distance, (iii) operating room within the mine design, (iv) availability and costs of power, fuel, etc., (v) weather / environment conditions, (vi) material type (rock, alluvium, etc.), (vii) mine life vs. capital required for a specific mining system, and (viii) operating characteristics of equipment.
- The size of equipment is influenced by a variety of factors, such as, mine design, production rates, comparative operating costs, and capital costs. The total number of units needed is determined from the required production rate and the productivity of individual unit. The outcome of equipment selection may dictate changes in mine design and rates of mining if the benefits of one type of mining system place it far ahead of other systems.

(a)

(b)

SELECTION OF EXCAVATOR AND TRANSPORTATION UNITS

To be practical; consideration of following three factors, to choose right size of a shovel or any single-bucket excavator plays an important role.

- Time factor
- Operational factor
- Bucket fill factor.

Time factor T_f *=Effective minutes available every hour/60*

It is the percentage availability of the equipment in unit time, which could be an hour or a shift

Operational factor (O_f*)*

This is a reflection of working conditions that includes layout, matching equipment meant for loading (mucking), transportation, crushing etc. This also takes into account the services in terms of lighting, ventilation (comforts), heat, humidity or any other factor that effects the performance of the equipment including the operator's efficiency who operates it. To apply corrections, usual guideline is mentioned below:

Conditions	Correction
Favorable	80%
Average	70%
Unfavorable	60% and below

Bucket fill factor (B_f)

It is the percentage of rated bucket capacity (m^3) to the one which will actually be delivered in a working cycle. It is based on the degree of fragmentations, or size of material to be filled in, and also bucket penetration, breakout force, bucket profile and ground engaging tools such as bucket teeth or replaceable cutting edge or lip

Table 17.2(a) Factors to be considered in the calculations.

Material	Fill factor
Loose material	
Mixed moist aggregates	95–100%
Uniform aggregates up to 3 mm	95–100%
3 to 9 mm	90–95%
12 mm and above	85–90%
Blasted rocks	
Well blasted	80–95%
Average	75–90%
Poor	60–75%
Others	
Rock dirt mixture	100–120%
Moist loam	100–110%
Soil, boulders, roots	80–100%
Cemented material	85–95%
Factor to be considered	
Favorable	120%
Average	90%
Unfavorable	60%

Table 17.2 (b) Output/hour[11] from a dipper shovel, as given by the manufacturers to choose the shovel based on the desired output.

Bucket capacity (m^3)	Rock output (m^3/hr.)	Earth bank output (m^3/hr.)
3.8	285–380	320–465
6.1	375–515	460–630
6.9	425–500	520–710
6.6	470–645	575–785
11.5	705–970	870–1185
19.1	1175–1585	1455–1910

Table 17.2(c) Truck size details.[11]

Normal Truck Size: 22, 30, 35, 40, 55, 85, 100, 130 tons (20, 27, 32, 36, 50, 77, 90, 117 tonnes)

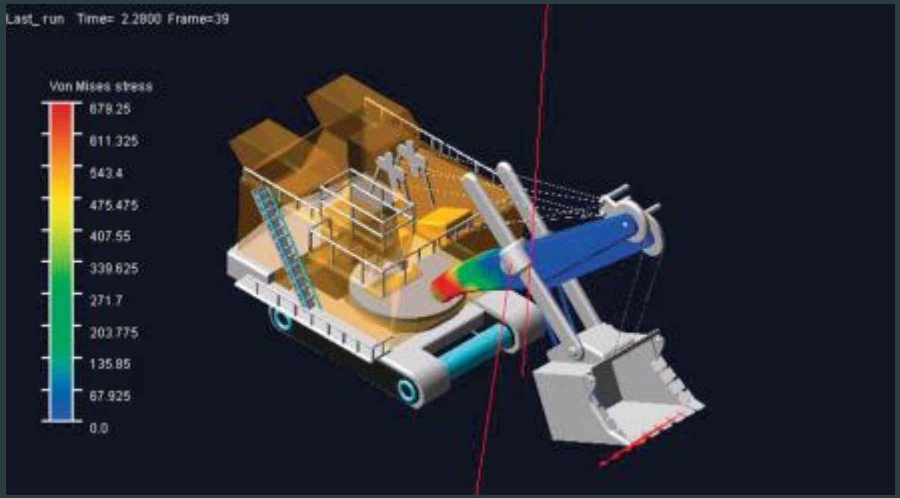
Giant size Trucks: 150, 175, 200, 250, 300, 350 tons. (135, 158, 180, 225, 270, 315 tonnes)

SELECTION OF SHOVEL/EXCAVATOR

Desired output/hr. = (Desired output in tons./shift)/Effective working hrs. per shift

$$OPD_h = OPD_s / W_h$$

Whereas: OPD_h – the desired output/hr.
OPD_s – the desires output/shift
W_h – Effective working hours/shift



Actual output to be achieved/hr. = (Desired output per hr.)/Time factor × Operational factor × Bucket fill factor

$$OPA_h = OPD_h / (TF \ X \ OF \ X \ BF) \qquad (17.20b)$$

Whereas: OPA_h – Actual output to achieved/hr.
TF – Time factor
OF – Operational factor
BF – Bucket fill factor

Actual output in m^3/hr. = Output to be achieved per hr. in tons./Density of loose or blasted rock.

$$OPA_{cum} = OPA_h / D_{loose} \qquad (17.20c)$$

Whereas: OPA_{cum} – Actual output to be achieved in m^3/hr.
D_{loose} – Density of the loose or blasted material in tons/m^3

Based on this calculation SELECT THE MATCHING SHOVEL/EXCAVATOR bucket capacity from the performance table supplied by the manufacturer (e.g. table 17.2(b)

Check! output per shift/1200 < bucket capacity selected (17.20d)[9]

Actual material/pass, in tons. = Bucket capacity in m^3 × Fill factor × Density of loose or (blasted) material

$$M_{PP} = B_{cap} \ X \ BF \ X \ D_{loose} \qquad (17.20e)$$

Whereas: M_{PP} – Material/pass of the excavator in tons.
B_{cap} – Bucket capacity in m^3.

TRUCK/DUMPER SELECTION

TRUCK CAPACITY = 4 or 5 × Bucket capacity of excavator in tons.

$$T_{cap} = 4 \text{ or } 5 \ X \ B_{capt} \quad (17.21a)$$

Whereas: T_{cap} – Truck capacity in tons.
B_{capt} – Bucket capacity of the excavator/shovel in tons.

Select truck of matching capacity from table 17.2(c).

Determine the cycle time of truck selected. The cycle time consists of the following operations:

- Loading time
- Travel time of the loaded truck from loading site to the discharge site
- Discharge or unloading time
- Travel time of empty truck
- Spot or change over time from one truck to another.

(*Continued*)

Once the cycle time is known or calculated, calculate number of trips by a truck/shift, using the relation:

Number of trips/shift/truck = (Effective time available per shift in minutes)/(cycle time in minutes)

$$NT_S = ETS_{min} / CYT_{min} \quad (17.21b)$$

$$OPT_S = Nt_s \times T_{cap} \quad (17.21c)$$

Whereas: OPT_S – Desired output per truck per shift in tons.
NT_S – Number of trips per shift per truck
T_{cap} – Truck capacity in tons.
ETS_{min} – Effective time available/shift in minutes
CYT_{min} – Cycle time in minutes

Number of trucks (NT) = (Desired Output/shift)/(Output/truck/shift) (17.21d)
or $NT = OPD_s / OPTS$

Calculation of loading time of shovel into trucks:
t_1 = (60 × Truck capacity in m^3)/Average of the Rated output of shovel per hr. in m^3 * (17.21e)
* – From the table supplied by the manufacture

Synchronization

$$\text{Check! cycle time, } CYT_{min} \le NT(t_l + t_s) \quad (17.21f)$$

Whereas: t_1 loading time of truck by the shovel or excavator, in minutes.
t_s – spot time in minutes.
NT – number of trucks.

17.5.11 THEORETICAL OUTPUT FROM AN EXCAVATOR/HR

$$O_{th} = 3600 \text{ x } E / T, \; m^3/hour \tag{17.22a}$$

O_{th} is the theoretical output/hour
E is bucket capacity in m^3 and T is cycle time in second.

In practice the factors such as bucket fill, time and bulking; need to be incorporated in the above mentioned formula and then it becomes:

$$O_{act} = 3600 \, E \, B_f \, T_f \, O_f \; / \, K \, T, \; m^3/hour \tag{17.22b}$$

Whereas: O_{act} – Actual output/hour;
B_f is bucket fill factor;
K is bulking factor;
T_f is time factor;
O_f – Operational factor.

Bulking factor (K) is the ratio of increase in volume of the material after getting fragmented to its volume in place or in-situ (original). It differs from material to material depending upon their densities.

Table 11.6. Swing correction factor. Atkinson (1992).

Angle of swing, degrees	45	60	75	90	120	150	180
Swing factor	1.2	1.1	1.05	1.00	0.91	0.84	0.77

Material	Density at the Borrow 10^3 (kg/m^3)	Bulking (Swell) Factor (%)
Basalt	2.4 - 3.1	75 - 80
Clay	1.8 - 2.6	20 - 40
Dolomite	2.8	50 - 60
Gneiss	2.69	75 - 80
Granite	2.6 - 2.8	75 - 80
Gravel, dry	1.80	20 - 30
Gravel, wet	2.00	20 - 30
Limestone	2.7 -2.8	75 - 80
Quartz	2.65	75 - 80
Rock		40 - 80
Sand, dry	1.60	20 - 30
Sand, wet	1.95	20 - 30
Sandstone	2.1 - 2.4	75 - 80
Slate	2.6 - 3.3	85 - 90
Soil	1.2 - 1.6	20 - 30

17.5.12 OUTPUT FROM A CONTINUOUS FLOW UNIT

$$O_{act} = 3600 \, E \, B_f \, T_f \, O_f \; / \, K \, S_s, \quad m^3/hour$$

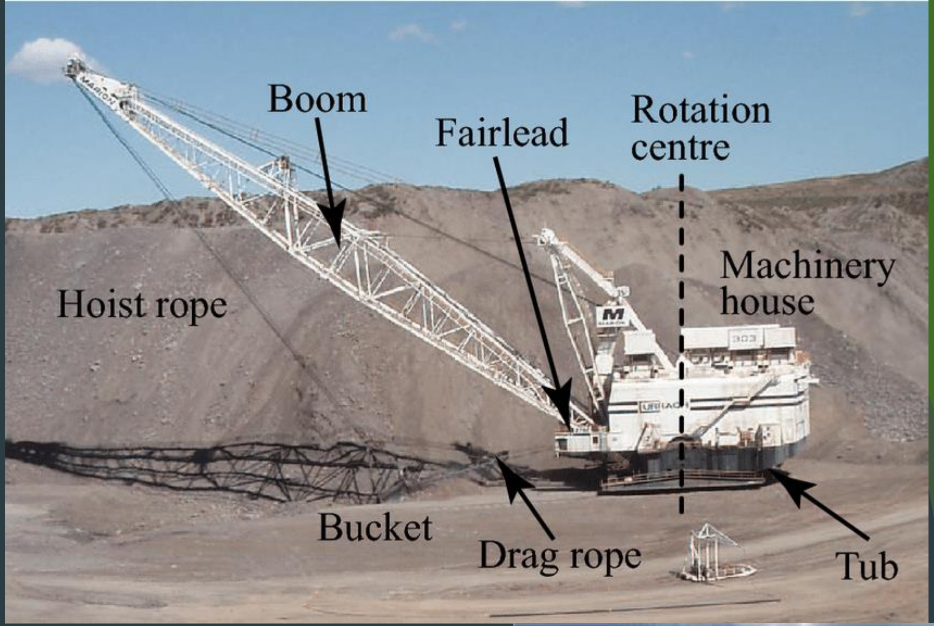
$$S_s = V_c \, N_b \, / \, \pi \, D$$

S_s is number of buckets discharged/sec;
V_c is cutting speed in m/sec;
N_b – Number of buckets in the wheel;
D – diameter of wheel in meters.





RWE
KRUPP


Boom
Fairlead
Rotation centre
Machinery house
Hoist rope
Bucket
Drag rope
Tub


CAT


216


CAT

11.9 SHOVEL PRODUCTIVITY EXAMPLE

To practice the concepts, calculate the production (bank volume per hour) from a shovel for which the following conditions apply:

- Dipper capacity (volume) = 15 yd^3
- Swing angle = 120° (time for a 90° swing is 30 sec.)
- Mechanical availability = 95%
- Working time = 50 min/hr.
- Optimum digging depth = 100% of optimum
- Swell = 30%
- Fill factor (fillability) = 90%
- Multi-bench mine
- Ore S.G. = 2.7

The calculation process will now be demonstrated:

Cycles/hour:

$$C = \text{theoretical cycles/hr} = \frac{60\text{ min/hr}}{0.5\text{ min/cy}} = 120\text{ cy/hr}$$

$$C_{BH} = 1.0$$

$$C_S = \text{swing correction } (120° \text{ rather than } 90°), \text{ therefore } C_S = 1.10$$

Thus, the corrected cycles = 120/1.1 = 109.09 cy/hr.

$$A = 0.95 \text{ Availability}$$

$$O = 50/60 = 0.83 \text{ working time}$$

Thus, corrected cycles = 109.09 × 0.95 × 0.83 = 86

$$P = \text{propel} = 0.85$$

Thus, actual cycles = 0.85 × 86 = 73.12 cy/hr.

Material/Cycle

$$Q_c = B_c \times F_f \times S_w$$

$$B_c = 15\text{ yd}^3 = \text{dipper capacity (yd}^3) - \text{“loose”}$$

$$F_f = \text{fill factor (expressed as a ratio)} = 0.90$$

Thus, material/cycle (loose) = 15 × 0.90 = 13.50 yd^3/cy

But the material has broken and swelled.

$$\%\text{ Swell} = 30\%$$

$$S_w = \frac{100}{100 + \%\text{Swell}} = \frac{100}{100 + 30} = 0.769$$

Thus,

$$\text{Material/cycle (bank} - \text{solid)} = 13.50/1.30 = 12.38\text{ yd}^3$$

Bank Volume/Hr

$$\text{Cycles/hr} \times \text{bank material/cycle} = 73.12 \times 12.38$$

$$= \underline{759 \text{ bank yd}^3\text{/hr}}$$

Tons/hr

$$\text{Tons/yd}^3 = \frac{2.7 \times 62.4 \times 27}{2000} = \frac{4949}{2000} = 2.27 \text{ tons/yd}^3$$

$$\text{Tons/hr} = 759 \times 2.27 = \underline{1723 \text{ tons/hr}}.$$

Tons/Shift = 8 hrs/shift × 1723 tons/hr = 13784 tons/shift for 15 yd^3 shovel.

Rule of Thumb = 1000 tons/yd^3 dipper capacity/8-hour shift.

In this case,

$$13784/15 = 919 \text{ tons/yd}^3 \text{ dipper capacity/shift}$$

which is similar to the rule-of-thumb value.

Thanks